

Network Design & Management (IT 3010)
3rd Year, 1st Semester

Assignment

Virtual Network Infrastructure Implementation using DNS, Mail Server and TCP Client-Server

Submitted to
Sri Lanka Institute of Information Technology

IT23687882

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

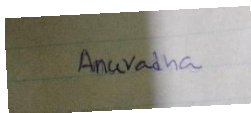
Declaration

I certify that this report does not incorporate without acknowledgement, any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where due reference is made in text.

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A rectangular piece of paper with the name "Anuvadha" written in blue ink.

Date: 12/04/2026

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1. Introduction

The current project considers the design and realization of a small network and its basic services, such as the connection between a client and a server. The main goal is to analyze the basics of configuring and functioning of some basic network services including DNS and mail server services using internet protocols.

An artificial network environment has been created based on the Linux operating system. Each participant has been provided with an individual IP addressing scheme according to the number of the student ID, which allowed creating a specific and unique network environment where some services, such as the domain name system and the mail server services, have been successfully configured.

In addition, a simple client-server application has been implemented in C language in order to demonstrate the principles of network communication.

Thus, the current work gives a detailed view of configuring networks and deploying services in networks.

2.Setting up the Virtual Environment

For the implementation of this task, a virtual computer environment was designed through virtualization software like VMware or VirtualBox. The purpose of the designed environment was to test different services in isolation from the actual computer network.

Process of Implementation:

The installation of a Linux OS (such as Ubuntu or CentOS) has been performed on the virtual machine.

Configuration of the virtual machine's IP address and other configuration settings for the virtual interfaces was done manually for the purpose of transferring data between the systems. Pinging was done to check for network connectivity.

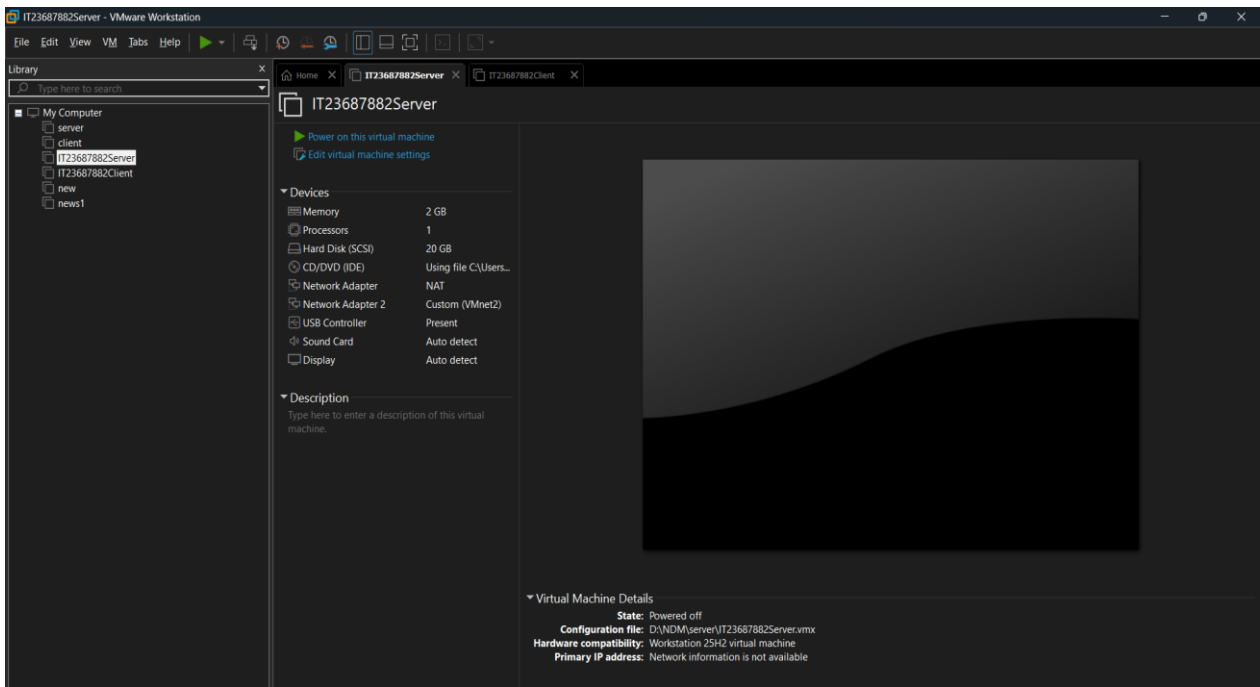


Figure 2.1: VMware virtual machine setup

2.1. Setting up the Network Adapter

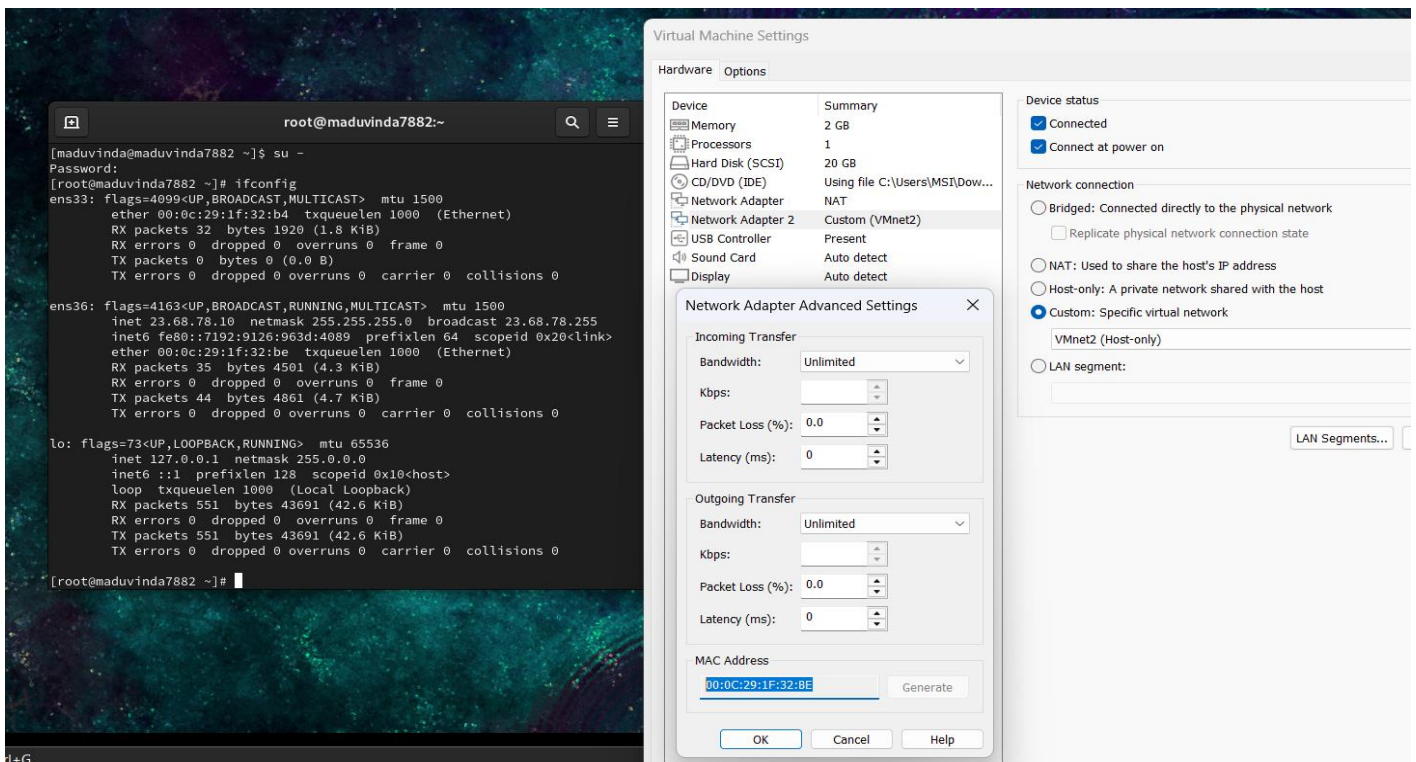
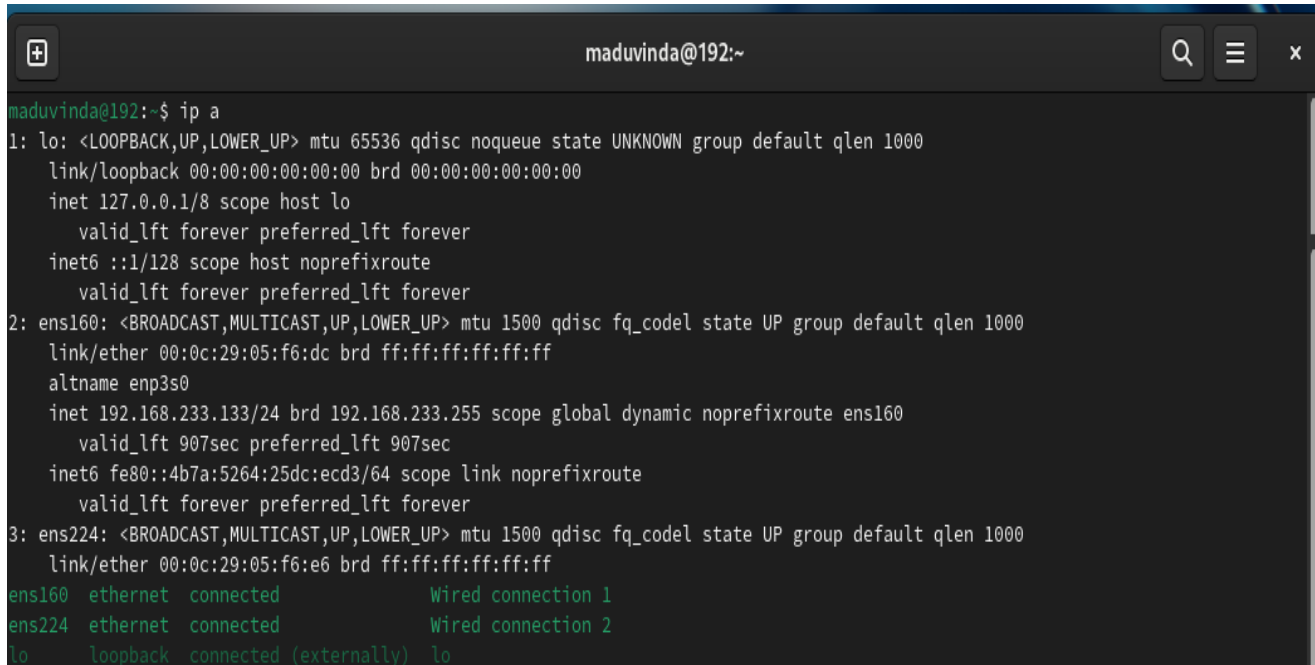


Figure 2.1.1: Network adapter configuration

The IP addressing scheme for the network was derived from the student registration number following a predefined format. Based on this, the virtual network adapter settings were configured to create a distinct and isolated network environment for the implementation.



```
maduvinda@192:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:05:f6:dc brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.233.133/24 brd 192.168.233.255 scope global dynamic noprefixroute ens160
        valid_lft 907sec preferred_lft 907sec
    inet6 fe80::4b7a:5264:25dc:ecd3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:05:f6:e6 brd ff:ff:ff:ff:ff:ff
ens160  ethernet  connected          Wired connection 1
ens224  ethernet  connected          Wired connection 2
lo      loopback  connected (externally) lo
```

Figure 2.1.2: IP configuration of client

VMnet2 virtual network was configured in order to allow communication among the various virtual machines operating in the system. The static IPs were assigned manually through the network configuration tool like nmcli. In order to ascertain whether connectivity was successful among the client and the server, ping test was carried out.

```

root@192:~
[root@192 ~]# nmcli device status
DEVICE  TYPE      STATE      CONNECTION
ens33   ethernet  connected  ens33
ens36   ethernet  connected  ens36
lo       loopback  connected (externally)  lo
[root@192 ~]#
[root@192 ~]# nmcli con mod ens36 ipv4.addresses 23.68.78.10/24
[root@192 ~]# nmcli con mod ens36 ipv4.method manual
[root@192 ~]# nmcli con up ens36
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
[root@192 ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1f:32:b4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.233.132/24 brd 192.168.233.255 scope global dynamic noprefixroute ens33
        valid_lft 941sec preferred_lft 941sec
    inet6 fe80::20c:29ff:fe1f:32b4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens36: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:1f:32:be brd ff:ff:ff:ff:ff:ff
    altname enp2s4
    inet 23.68.78.10/24 brd 23.68.78.255 scope global noprefixroute ens36
        valid_lft forever preferred_lft forever
    inet6 fe80::7192:9126:963d:4085/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@192 ~]#

```

```

[root@192 ~]# nmcli con modify ens36 ipv4.gateway 23.68.78.1
[root@192 ~]# nmcli con modify ens36 ipv4.dns 23.68.78.10
[root@192 ~]# nmcli con up ens36
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
[root@192 ~]#

```

Figure 2.1.3: Ping test between client and server using nmcli

```

maduvinda@192:~$ nmcli con show
NAME                UUID                                  TYPE      DEVICE
lo                  c6b5e376-3742-4479-adfd-350ed64914b2  loopback  lo
Wired connection 2  69172438-dbc1-35fd-aa3d-b152d91fca9e  ethernet  ens224
Wired connection 1  bbf9dd4b-3714-30f6-b6e3-fcb9e96c248b  ethernet  --
maduvinda@192:~$

maduvinda@192:~$
maduvinda@192:~$ ping -c 4 23.68.78.10
PING 23.68.78.10 (23.68.78.10) 56(84) bytes of data.
64 bytes from 23.68.78.10: icmp_seq=1 ttl=64 time=6.70 ms
64 bytes from 23.68.78.10: icmp_seq=2 ttl=64 time=5.56 ms
64 bytes from 23.68.78.10: icmp_seq=3 ttl=64 time=6.85 ms
^C
--- 23.68.78.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 5.556/6.369/6.851/0.578 ms
maduvinda@192:~$

```

Figure 2.1.4: Network connectivity test with nmcli and ping.

2.2. Configuring the System Hostname

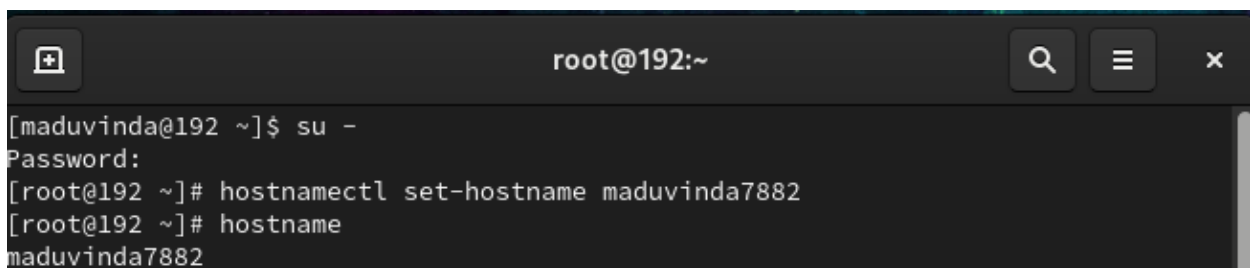
A hostname was assigned to the system using the format <maduvinda><7882>, which helps in identifying the machine within the network environment.

Implementation

- The hostname was configured using the command:
hostnamectl set-hostname maduvinda7882

Verification

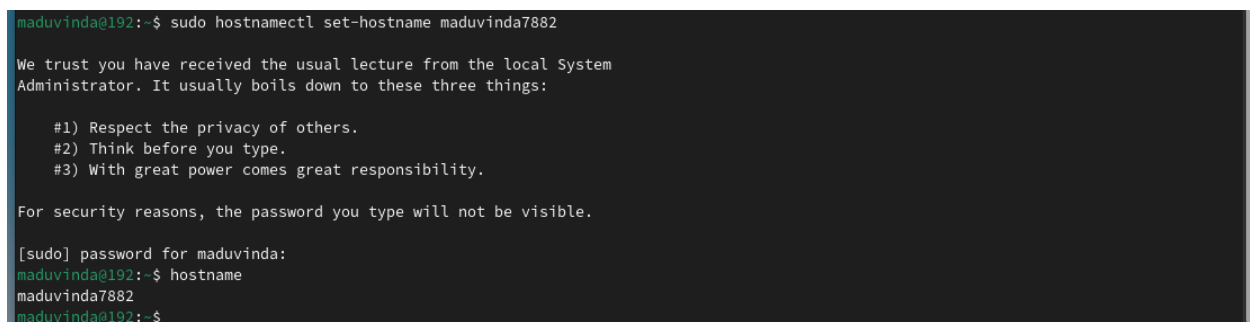
- The configured hostname was verified successfully as:
maduvinda7882

A terminal window titled 'root@192:~' with search, menu, and close icons in the top right. The terminal shows a user switching from 'maduvinda@192 ~' to 'root@192 ~' using 'su -'. Then, the command 'hostnamectl set-hostname maduvinda7882' is executed, followed by 'hostname', which outputs 'maduvinda7882'.

```
[maduvinda@192 ~]$ su -
Password:
[root@192 ~]# hostnamectl set-hostname maduvinda7882
[root@192 ~]# hostname
maduvinda7882
```

Figure 2.2.1: Server hostname

Client: fedora

A terminal window showing the execution of 'sudo hostnamectl set-hostname maduvinda7882' on a client. It displays a warning about system administrator responsibilities, a security note about password visibility, and the successful execution of the command, resulting in the hostname 'maduvinda7882'.

```
maduvinda@192:~$ sudo hostnamectl set-hostname maduvinda7882

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for maduvinda:
maduvinda@192:~$ hostname
maduvinda7882
maduvinda@192:~$
```

Figure 2.2.2: Client hostname


```

[root@192 ~]# nmtui
[root@192 ~]# ping google.com
PING google.com (172.217.194.102) 56(84) bytes of data.
64 bytes from si-in-f102.1e100.net (172.217.194.102): icmp_seq=1 ttl=128 time=87.2 ms
64 bytes from 102.194.217.172.in-addr.arpa (172.217.194.102): icmp_seq=2 ttl=128 time=86.6 ms
64 bytes from 102.194.217.172.in-addr.arpa (172.217.194.102): icmp_seq=3 ttl=128 time=84.3 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 84.256/86.021/87.237/1.277 ms
[root@192 ~]# nmtui
[root@192 ~]# ping 23.68.78.10
PING 23.68.78.10 (23.68.78.10) 56(84) bytes of data.
64 bytes from 23.68.78.10: icmp_seq=1 ttl=64 time=0.134 ms
64 bytes from 23.68.78.10: icmp_seq=2 ttl=64 time=0.217 ms
64 bytes from 23.68.78.10: icmp_seq=3 ttl=64 time=0.205 ms
^C
--- 23.68.78.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 0.134/0.185/0.217/0.036 ms
[root@192 ~]# nmtui
[root@192 ~]#

```

Figure Connectivity verification using ping to Google domain and external IP address(server side)(23.68.78.10).

```

maduvinda@192:~$ ping 23.68.78.10
PING 23.68.78.10 (23.68.78.10) 56(84) bytes of data.
64 bytes from 23.68.78.10: icmp_seq=1 ttl=64 time=4.38 ms
64 bytes from 23.68.78.10: icmp_seq=2 ttl=64 time=1.44 ms
64 bytes from 23.68.78.10: icmp_seq=3 ttl=64 time=2.51 ms
^C
--- 23.68.78.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.442/2.777/4.375/1.211 ms
maduvinda@192:~$ ping google.com
PING google.com (142.251.10.102) 56(84) bytes of data.
64 bytes from sd-in-f102.1e100.net (142.251.10.102): icmp_seq=1 ttl=128 time=63.8 ms
64 bytes from sd-in-f102.1e100.net (142.251.10.102): icmp_seq=2 ttl=128 time=59.8 ms
64 bytes from sd-in-f102.1e100.net (142.251.10.102): icmp_seq=3 ttl=128 time=58.8 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 58.821/60.784/63.763/2.141 ms
maduvinda@192:~$

```

Figure Connectivity verification using ping to Google domain and external IP address(client side)(23.68.78.10).

3.Implementation of the System

3.1. Setting up the DNS Server

Domain Name Service (DNS) server was also configured in the virtual machine in order to resolve the domain names in the network. This domain name was selected according to the student ID (e.g. IT23687882.lk). Testing of DNS services was done for proper mapping of domain names with the IP address.

DNS Installation

Installation of DNS server packages can be done by using the below mentioned command.

dnf install bind bind-utils -y

```
root@192:~  
[root@192 ~]# sudo dnf install bind bind-utils -y  
CentOS Stream 9 - BaseOS 3.2 MB/s | 8.9 MB 00:02  
CentOS Stream 9 - AppStream 3.3 MB/s | 27 MB 00:08  
CentOS Stream 9 - Extras packages 11 kB/s | 21 kB 00:01  
Package bind-utils-32:9.16.23-4.el9.x86_64 is already installed.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
bind	x86_64	32:9.16.23-40.el9	appstream	505 k
Upgrading:				
bind-libs	x86_64	32:9.16.23-40.el9	appstream	1.2 M
bind-license	noarch	32:9.16.23-40.el9	appstream	13 k
bind-utils	x86_64	32:9.16.23-40.el9	appstream	206 k
openssl	x86_64	1:3.5.5-1.el9	baseos	1.5 M
openssl-libs	x86_64	1:3.5.5-1.el9	baseos	2.3 M
Installing dependencies:				
bind-dnssec-doc	noarch	32:9.16.23-40.el9	appstream	45 k
openssl-fips-provider	x86_64	1:3.5.5-1.el9	baseos	814 k
python3-bind	noarch	32:9.16.23-40.el9	appstream	67 k
python3-ply	noarch	3.11-14.el9	baseos	106 k
Installing weak dependencies:				
bind-dnssec-utils	x86_64	32:9.16.23-40.el9	appstream	116 k

```
=====
```

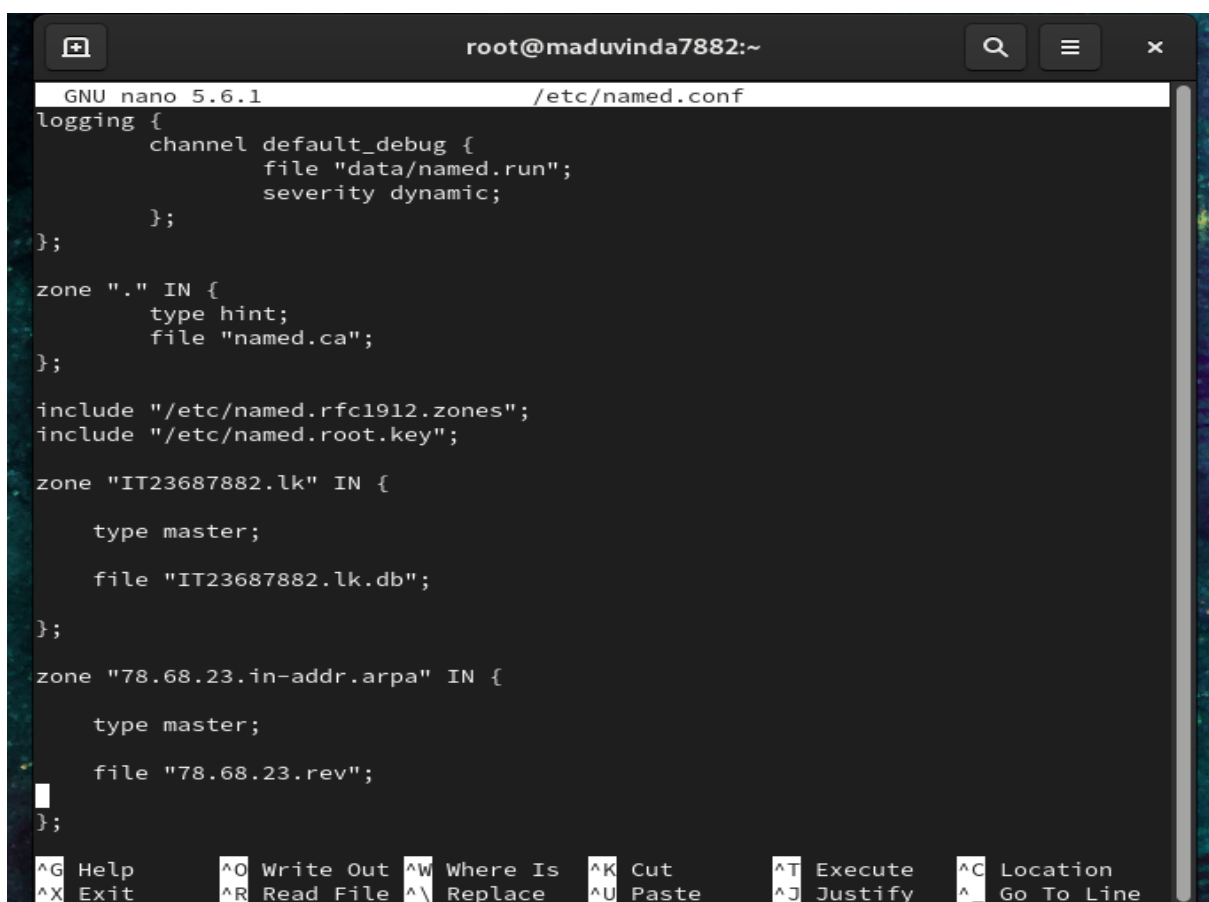
Transaction Summary			
Install	6 Packages		
Upgrade	5 Packages		
Total download size: 6.9 M			
Downloading Packages:			
CentOS Stream 9 - BaseOS	174%	[=====]	7
CentOS Stream 9 - AppStream	174%	[=====]	7
(1/11): python3-ply-3.11-14.el9.noarch.rpm	71 kB/s	106 kB	00:01
(2/11): bind-9.16.23-40.el9.x86_64.rpm	278 kB/s	505 kB	00:01
(3/11): bind-dnssec-utils-9.16.23-40.el9.x86_64.rpm	422 kB/s	116 kB	00:00
(4/11): python3-bind-9.16.23-40.el9.noarch.rpm	243 kB/s	67 kB	00:00
(5/11): bind-dnssec-doc-9.16.23-40.el9.noarch.rpm	43 kB/s	45 kB	00:01
(6/11): openssl-3.5.5-1.el9.x86_64.rpm	955 kB/s	1.5 MB	00:01
(7/11): openssl-fips-provider-3.5.5-1.el9.x86_64.rpm	185 kB/s	814 kB	00:04
(8/11): bind-license-9.16.23-40.el9.noarch.rpm	48 kB/s	13 kB	00:00
(9/11): bind-libs-9.16.23-40.el9.x86_64.rpm	963 kB/s	1.2 MB	00:01
(10/11): bind-utils-9.16.23-40.el9.x86_64.rpm	196 kB/s	206 kB	00:01
(11/11): openssl-libs-3.5.5-1.el9.x86_64.rpm	387 kB/s	2.3 MB	00:06
Total			
CentOS Stream 9 - BaseOS	698 kB/s	6.9 MB	00:10
	1.1 MB/s	1.6 kB	00:00

```
root@192:~  
Total 698 kB/s | 6.9 MB 00:10  
CentOS Stream 9 - BaseOS 1.1 MB/s | 1.6 kB 00:00  
Importing GPG key 0x8483C65D:  
  Userid : "CentOS (CentOS Official Signing Key) <security@centos.org>"  
  Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D  
  From : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial  
Key imported successfully  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing : 1/16  
  Upgrading : bind-license-32:9.16.23-40.el9.noarch 1/16  
  Installing : bind-dnssec-doc-32:9.16.23-40.el9.noarch 2/16  
  Upgrading : openssl-libs-1:3.5.5-1.el9.x86_64 3/16  
  Installing : openssl-fips-provider-1:3.5.5-1.el9.x86_64 4/16  
  Upgrading : bind-libs-32:9.16.23-40.el9.x86_64 5/16  
  Upgrading : bind-utils-32:9.16.23-40.el9.x86_64 6/16  
  Installing : python3-ply-3.11-14.el9.noarch 7/16  
  Installing : python3-bind-32:9.16.23-40.el9.noarch 8/16  
  Installing : bind-dnssec-utils-32:9.16.23-40.el9.x86_64 9/16  
  Running scriptlet: bind-32:9.16.23-40.el9.x86_64 10/16  
  Installing : bind-32:9.16.23-40.el9.x86_64 10/16  
  Running scriptlet: bind-32:9.16.23-40.el9.x86_64 10/16  
  Upgrading : openssl-1:3.5.5-1.el9.x86_64 11/16  
  Cleanup : openssl-1:3.2.2-6.el9.x86_64 12/16  
  Cleanup : bind-utils-32:9.16.23-40.el9.x86_64 13/16  
  Cleanup : bind-libs-32:9.16.23-40.el9.x86_64 14/16  
  Cleanup : bind-license-32:9.16.23-40.el9.noarch 15/16  
  Cleanup : openssl-libs-1:3.2.2-6.el9.x86_64 16/16  
  Running scriptlet: openssl-libs-1:3.2.2-6.el9.x86_64 16/16  
  Verifying : openssl-fips-provider-1:3.5.5-1.el9.x86_64 1/16  
  Verifying : python3-ply-3.11-14.el9.noarch 2/16  
  Verifying : bind-32:9.16.23-40.el9.x86_64 3/16  
  Verifying : bind-dnssec-doc-32:9.16.23-40.el9.noarch 4/16  
  Verifying : bind-dnssec-utils-32:9.16.23-40.el9.x86_64 5/16  
  Verifying : python3-bind-32:9.16.23-40.el9.noarch 6/16  
  Verifying : openssl-1:3.5.5-1.el9.x86_64 7/16  
  Verifying : openssl-1:3.2.2-6.el9.x86_64 8/16  
  Verifying : openssl-libs-1:3.5.5-1.el9.x86_64 9/16  
  Verifying : openssl-libs-1:3.2.2-6.el9.x86_64 10/16  
  Verifying : bind-libs-32:9.16.23-40.el9.x86_64 11/16  
  Verifying : bind-libs-32:9.16.23-40.el9.x86_64 12/16  
  Verifying : bind-license-32:9.16.23-40.el9.noarch 13/16  
  Verifying : bind-utils-32:9.16.23-40.el9.x86_64 14/16  
  Verifying : bind-utils-32:9.16.23-40.el9.x86_64 15/16  
  Verifying : bind-utils-32:9.16.23-40.el9.x86_64 16/16
```

```
Upgraded:
bind-libs-32:9.16.23-40.el9.x86_64      bind-license-32:9.16.23-40.el9.noarch      bind-utils-32:9.16.23-40.el9.x86_64
openssl-1:3.5.5-1.el9.x86_64            openssl-libs-1:3.5.5-1.el9.x86_64
Installed:
bind-32:9.16.23-40.el9.x86_64            bind-dnssec-doc-32:9.16.23-40.el9.noarch
bind-dnssec-utils-32:9.16.23-40.el9.x86_64  openssl-fips-provider-1:3.5.5-1.el9.x86_64
python3-bind-32:9.16.23-40.el9.noarch      python3-ply-3.11-14.el9.noarch
Complete!
[root@192 ~]#
```

Figure 3.1.1: BIND installation verification

Configuration:



```
root@maduvinda7882:~
GNU nano 5.6.1 /etc/named.conf
logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

zone "IT23687882.lk" IN {
    type master;

    file "IT23687882.lk.db";
};

zone "78.68.23.in-addr.arpa" IN {
    type master;

    file "78.68.23.rev";
};

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Figure 3.1.2: named.conf configuration

Forward Zone

The zone file's setup for the zone of the forward lookup zone of the domain name IT23687882.lk has been validated since it provides forward name resolution using domain names and individual names with IP addresses within the network. A record entry has been created in the zone file for resolving the important host names like *ns*, *mail*, *www*, and *client* to the corresponding IP address. MX entry has been set up which will define the mail server for performing mail services in the domain.

Post configuration, the zone file has been verified for any errors, and the DNS service was restarted.



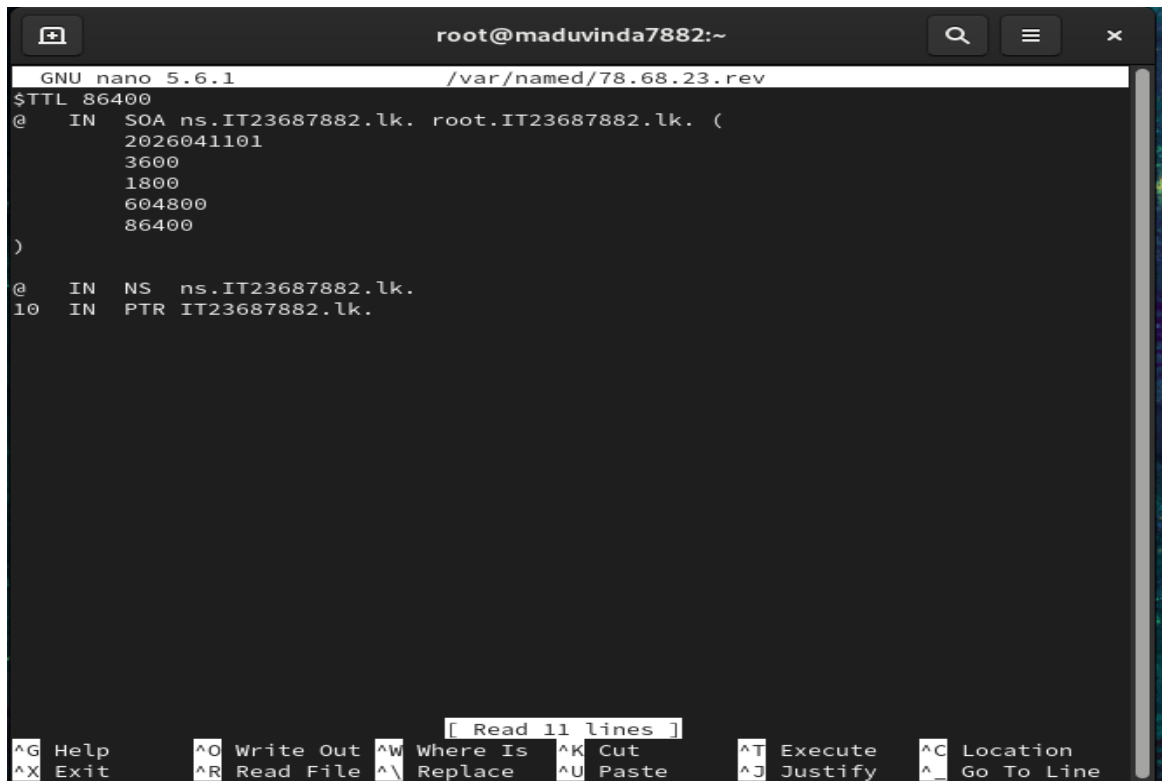
```
root@maduvinda7882:~  
GNU nano 5.6.1 /var/named/IT23687882.lk.db  
$TTL 86400  
@ IN SOA ns.IT23687882.lk. root.IT23687882.lk. (  
    2026041101  
    3600  
    1800  
    604800  
    86400 )  
@ IN NS      ns.IT23687882.lk.  
ns IN A      23.68.78.10  
@ IN A      23.68.78.10  
www IN A     23.68.78.10  
mail IN A    23.68.78.10  
@ IN MX 10   mail.IT23687882.lk.  
[ Read 13 lines ]  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Figure 3.1.3: Forward zone configuration

Reverse Zone

To allow users to access their devices without having to remember long strings of numbers (IP Address) by providing them with a simple way to know what device they are trying to reach (HostName) this is accomplished using Reverse Lookup Zones. The Reverse Lookup Zone provides IP to Host Name resolution using PTR records - which maps an IP Address back to its associated Host Name - *ns*, *mail*, *client*.

Prior to restarting the DNS Service, a Check was performed on the Reverse Zone File to ensure that all entries were accurate; then the DNS service was restarted. Following the restart of the DNS Service, testing was performed to ensure that all the IP Addresses were correctly resolved back to their respective Host Names.



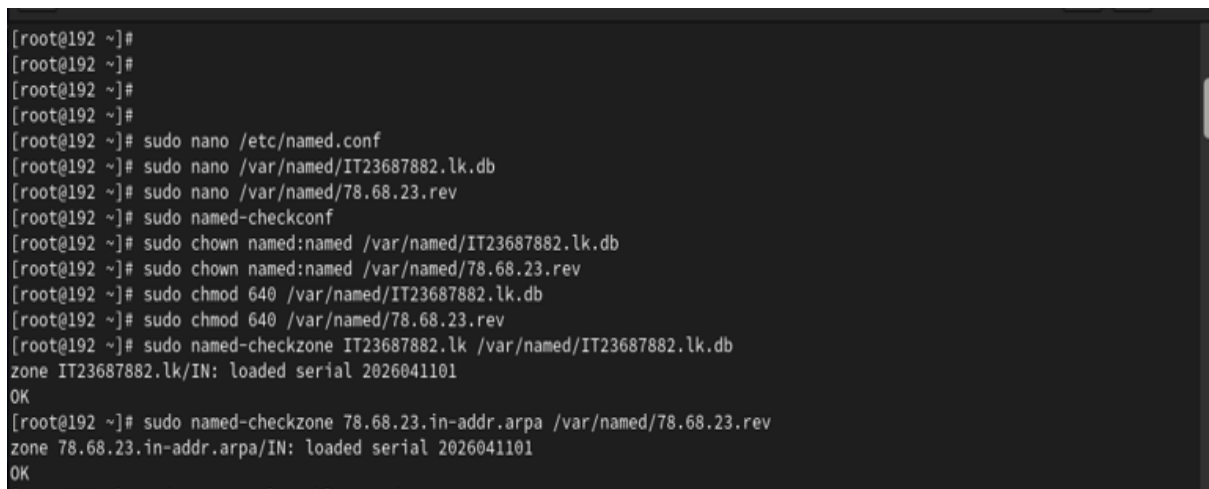
The screenshot shows a terminal window with the title bar "root@maduvinda7882:~". The terminal is running the GNU nano 5.6.1 editor, editing the file /var/named/78.68.23.rev. The content of the file is as follows:

```
STTL 86400
@ IN SOA ns.IT23687882.lk. root.IT23687882.lk. (
    2026041101
    3600
    1800
    604800
    86400
)

@ IN NS ns.IT23687882.lk.
10 IN PTR IT23687882.lk.
```

The bottom of the terminal shows the nano editor's status bar with various keyboard shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line. A message "[Read 11 lines]" is also visible.

Figure 3.1.4: Reverse zone configuration



The screenshot shows a terminal window with the title bar "[root@192 ~]#". The terminal displays the following commands and their outputs:

```
[root@192 ~]#
[root@192 ~]#
[root@192 ~]#
[root@192 ~]#
[root@192 ~]# sudo nano /etc/named.conf
[root@192 ~]# sudo nano /var/named/IT23687882.lk.db
[root@192 ~]# sudo nano /var/named/78.68.23.rev
[root@192 ~]# sudo named-checkconf
[root@192 ~]# sudo chown named:named /var/named/IT23687882.lk.db
[root@192 ~]# sudo chown named:named /var/named/78.68.23.rev
[root@192 ~]# sudo chmod 640 /var/named/IT23687882.lk.db
[root@192 ~]# sudo chmod 640 /var/named/78.68.23.rev
[root@192 ~]# sudo named-checkzone IT23687882.lk /var/named/IT23687882.lk.db
zone IT23687882.lk/IN: loaded serial 2026041101
OK
[root@192 ~]# sudo named-checkzone 78.68.23.in-addr.arpa /var/named/78.68.23.rev
zone 78.68.23.in-addr.arpa/IN: loaded serial 2026041101
OK
```

Figure 3.1.5: Named-check configuration and named-check zone configuration

dig @127.0.0.1 IT23687882.lk MX

dig @127.0.0.1 portal.IT23687882.lk CNAME

3.2. Setting up the Mail Server

An internal email setup of a mail server and mail client was created to facilitate communication over a network within a virtual machine. This setup achieved user-to-user email transfer capabilities within the network.

Installation

dnf install postfix s-nail-y

Configuration/Testing

- Postfix service was configured to perform mail transfer function on the local machine
- Test mail users were created in the local system for testing
- Email messages were successfully sent/received between test users via the mail client
- Successful delivery to test users indicates that the mail server ran without errors.

```
root@192:~  
[root@192 ~]# dnf install postfix s-nail -y  
Last metadata expiration check: 1:06:24 ago on Sun 12 Apr 2026 12:47:26 AM +0530.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
postfix	x86_64	2:3.5.25-2.el9	appstream	1.5 M
s-nail	x86_64	14.9.22-9.el9	appstream	620 k

```
=====
```

Transaction Summary

=====

Install 2 Packages

Total download size: 2.1 M
Installed size: 5.6 M
Downloading Packages:
(1/2): s-nail-14.9.22-9.el9.x86_64.rpm 615 kB/s | 620 kB 00:01
(2/2): postfix-3.5.25-2.el9.x86_64.rpm 444 kB/s | 1.5 MB 00:03

Total	540 kB/s	2.1 MB	00:04
-------	----------	--------	-------

```
-----  
Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing      :  
  Running scriptlet: s-nail-14.9.22-9.el9.x86_64 1/1  
  Installing     : s-nail-14.9.22-9.el9.x86_64 1/2  
  Running scriptlet: s-nail-14.9.22-9.el9.x86_64 1/2  
  Installing     : postfix-2:3.5.25-2.el9.x86_64 2/2  
  Running scriptlet: postfix-2:3.5.25-2.el9.x86_64 2/2  
  Installing     : postfix-2:3.5.25-2.el9.x86_64 2/2  
  Running scriptlet: postfix-2:3.5.25-2.el9.x86_64 2/2  
  Verifying      : postfix-2:3.5.25-2.el9.x86_64 1/2  
  Verifying      : s-nail-14.9.22-9.el9.x86_64 2/2  
Installed:  
postfix-2:3.5.25-2.el9.x86_64 s-nail-14.9.22-9.el9.x86_64  
Complete!  
[root@192 ~]#
```

Figure 3.2.1: Postfix installation

Configuration Process

The /etc/postfix/main.cf configuration file for Postfix has been edited, creating a mail server with the following key parameters:

Configuration:

- myhostname = mail.IT23687882.lk
- mydomain = IT23687882.lk
- myorigin = \$mydomain
- inet_interfaces = all
- inet_protocols = ipv4
- mydestination = \$myhostname, localhost.\$mydomain, localhost, \$mydomain
- mynetworks = 127.0.0.0/8, 23.68.78.0/24
- home_mailbox = Maildir/

Using these settings, a mail server was created that handles internal email communication within the IT23687882.lk domain.


```

root@maduvinda7882:~
GNU nano 5.6.1 /etc/postfix/main.cf

# Directory with PEM format Certification Authority certificates that the
# Postfix SMTP client uses to verify a remote SMTP server certificate.
#
smtp_tls_CApath = /etc/pki/tls/certs

# The full pathname of a file containing CA certificates of root CAs
# trusted to sign either remote SMTP server certificates or intermediate CA
# certificates.
#
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt

# Use TLS if this is supported by the remote SMTP server, otherwise use
# plaintext (opportunistic TLS outbound).
#
smtp_tls_security_level = may
meta_directory = /etc/postfix
shlib_directory = /usr/lib64/postfix

myhostname = mail.IT23687882.lk
mydomain = IT23687882.lk
myorigin = $mydomain
inet_interfaces = all
inet_protocols = ipv4
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mynetworks = 127.0.0.0/8, 23.68.78.0/24
home_mailbox = Maildir/

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line

```

Figure 3.2.2: Postfix main.cf configuration

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```

[root@192 ~]# sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-04-12 02:15:30 +0530; 8min ago
     Process: 36780 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)
     Process: 36781 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
     Process: 36785 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
     Process: 36786 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
   Main PID: 36854 (master)
     Tasks: 3 (Limit: 10717)
    Memory: 6.2M
       CPU: 327ms
    CGroup: /system.slice/postfix.service
            └─36854 /usr/libexec/postfix/master -w
              └─36855 pickup -l -t unix -u
                └─36856 qmgr -l -t unix -u

Apr 12 02:15:30 maduvinda7882 postfix/master[36854]: warning: /etc/postfix/main.cf, line 746: overriding earlier entry: inet_protocols=all
Apr 12 02:15:30 maduvinda7882 postfix/master[36854]: warning: /etc/postfix/main.cf, line 747: overriding earlier entry: mydestination=$myhostname, localhost.$mydo
Apr 12 02:15:30 maduvinda7882 postfix/master[36854]: daemon started -- version 3.5.25, configuration /etc/postfix
Apr 12 02:15:30 maduvinda7882 systemd[1]: Started Postfix Mail Transport Agent.
Apr 12 02:15:30 maduvinda7882 postfix/qmgr[36856]: warning: /etc/postfix/main.cf, line 745: overriding earlier entry: inet_interfaces=localhost
Apr 12 02:15:30 maduvinda7882 postfix/qmgr[36856]: warning: /etc/postfix/main.cf, line 746: overriding earlier entry: inet_protocols=all
Apr 12 02:15:30 maduvinda7882 postfix/qmgr[36856]: warning: /etc/postfix/main.cf, line 747: overriding earlier entry: mydestination=$myhostname, localhost.$mydo
Apr 12 02:15:30 maduvinda7882 postfix/pickup[36855]: warning: /etc/postfix/main.cf, line 745: overriding earlier entry: inet_interfaces=localhost
Apr 12 02:15:30 maduvinda7882 postfix/pickup[36855]: warning: /etc/postfix/main.cf, line 746: overriding earlier entry: inet_protocols=all
Apr 12 02:15:30 maduvinda7882 postfix/pickup[36855]: warning: /etc/postfix/main.cf, line 747: overriding earlier entry: mydestination=$myhostname, localhost.$mydo
lines 1-26/26 (END)

[root@192 ~]# sudo firewall-cmd --permanent --add-service=smtp

```

Figure 3.2.3: Activation of Mail server

User Setup:

- Created users: user1, user2
- Created Maildir directories

```
root@192:~  
success  
[root@192 ~]# sudo nano /etc/postfix/main.cf  
[root@192 ~]# sudo useradd user1  
[root@192 ~]# sudo passwd user1  
Changing password for user user1.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@192 ~]# sudo useradd user2  
[root@192 ~]# sudo passwd user2  
Changing password for user user2.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@192 ~]# sudo mkdir -p /home/user1/Maildir
```

Figure 3.2.4: Creating User1 and User2 Password

```
[root@maduvinda7882 ~]# sudo mkdir -p /home/user1/Maildir/{cur,new,tmp}  
[root@maduvinda7882 ~]# sudo chown -R user1:user1 /home/user1/Maildir  
[root@maduvinda7882 ~]# exit  
logout  
[user1@maduvinda7882 ~]$ exit  
logout
```

Figure 3.2.5: Creating Maildir for user1

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```
[root@maduvinda7882 ~]# exit  
logout  
[maduvinda@maduvinda7882 ~]$ sudo mkdir -p /home/test  
[sudo] password for maduvinda:  
maduvinda is not in the sudoers file. This incident will be reported.  
[maduvinda@maduvinda7882 ~]$ su - maduvinda  
Password:  
[maduvinda@maduvinda7882 ~]$ sudo mkdir -p /home/test
```

Figure 3.2.6: Linux user privilege check with sudo and su commands

```
maduvinda@maduvinda7882 ~]$  
maduvinda@maduvinda7882 ~]$ sudo mkdir -p /home/user2/Maildir  
maduvinda@maduvinda7882 ~]$ sudo mkdir -p /home/user2/Maildir  
maduvinda@maduvinda7882 ~]$ sudo mkdir -p /home/user2/Maildir/{cur,new,tmp}  
maduvinda@maduvinda7882 ~]$ sudo chown -R user2:user2 /home/user2/Maildir  
maduvinda@maduvinda7882 ~]$ sudo systemctl enable postfix  
maduvinda@maduvinda7882 ~]$ sudo systemctl restart postfix
```

Figure 3.2.7: Creating Maildir for user2

```
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service
```

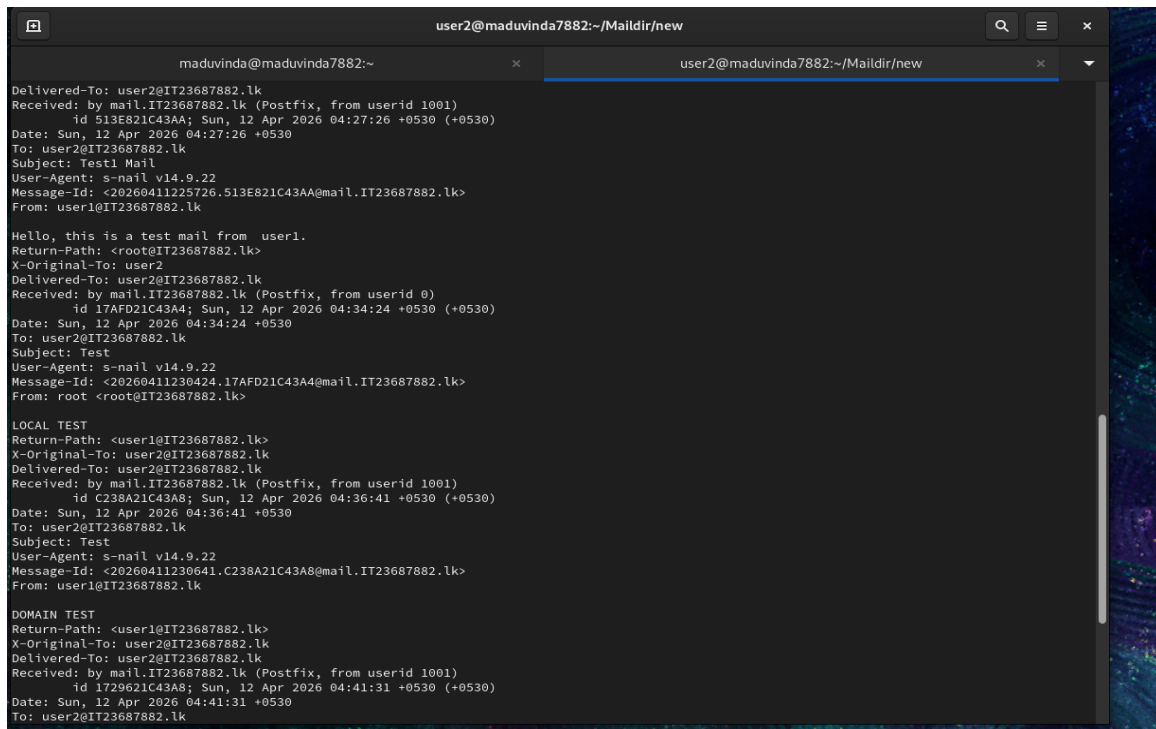
Figure 3.2.8: Postfix service enabled with systemd symlink creation.

```
maduvinda@maduvinda7882 ~]$ sudo firewall-cmd --permanent --add-service=smtp
[sudo] password for maduvinda:
sudo: firewall-cmd: command not found
maduvinda@maduvinda7882 ~]$ sudo firewall-cmd --permanent --add-service=smtp
warning: ALREADY_ENABLED: smtp
success
maduvinda@maduvinda7882 ~]$ sudo firewall-cmd --reload
success
```

Figure 3.2.9: Firewall configuration with firewall-cmd enabling SMTP service and reload.

```
[user2@maduvinda7882 new]$ su - user1
Password:
[user1@maduvinda7882 ~]$ echo "Hello, this is a test mail from user1." | mail -s "Test1 Mail" user2@IT23687882.lk
sendmail: warning: /etc/postfix/main.cf, line 745: overriding earlier entry: inet_interfaces=localhost
sendmail: warning: /etc/postfix/main.cf, line 746: overriding earlier entry: inet_protocols=all
sendmail: warning: /etc/postfix/main.cf, line 747: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
postdrop: warning: /etc/postfix/main.cf, line 745: overriding earlier entry: inet_interfaces=localhost
postdrop: warning: /etc/postfix/main.cf, line 746: overriding earlier entry: inet_protocols=all
postdrop: warning: /etc/postfix/main.cf, line 747: overriding earlier entry: mydestination=$myhostname, localhost.$mydomain, localhost
[user1@maduvinda7882 ~]$ su - user2
Password:
[user2@maduvinda7882 ~]$ cd Maildir/new
[user2@maduvinda7882 new]$ ls
1775948246.Vfd00111ab87M349452.maduvinda7882  1775948801.Vfd00111ab8cM833121.maduvinda7882
1775948664.Vfd00111ab83M119775.maduvinda7882  1775949091.Vfd00111ab8dM114525.maduvinda7882
[user2@maduvinda7882 new]$
```

Figure : Email received and verified using Maildir



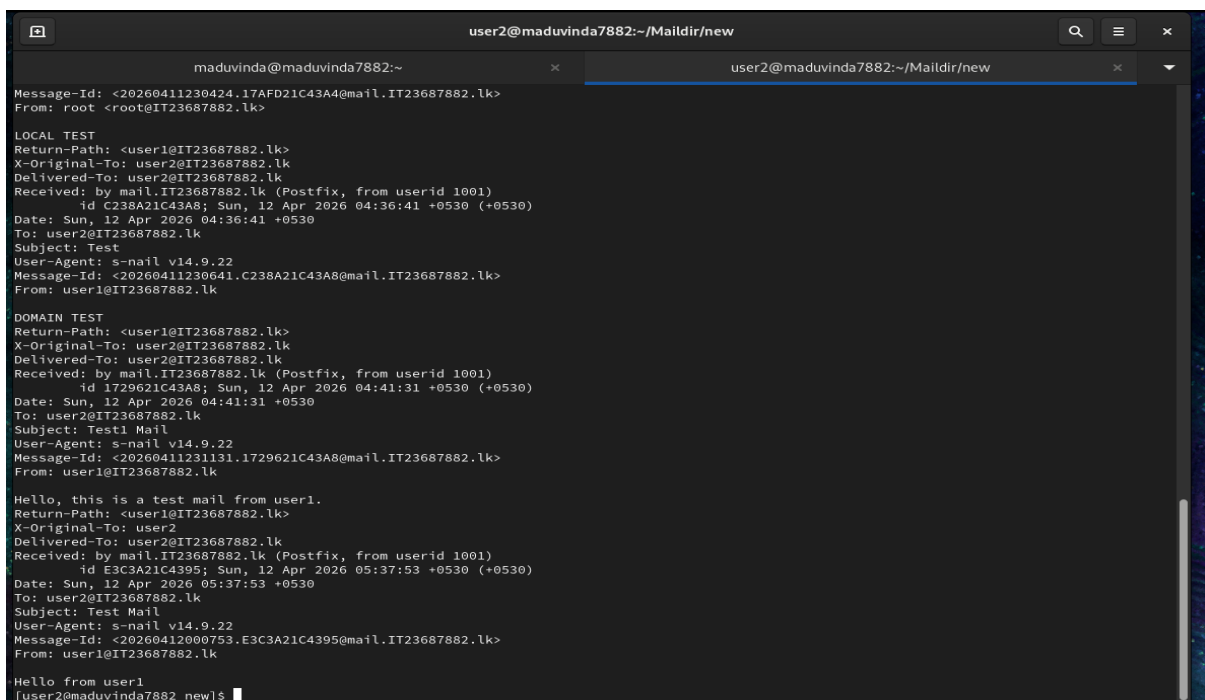
```
user2@maduvinda7882:~/Maildir/new
maduvinda@maduvinda7882:~
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id 513E821C43AA; Sun, 12 Apr 2026 04:27:26 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:27:26 +0530
To: user2@IT23687882.lk
Subject: Test1 Mail
User-Agent: s-nail v14.9.22
Message-Id: <20260411225726.513E821C43AA@mail.IT23687882.lk>
From: user1@IT23687882.lk

Hello, this is a test mail from user1.
Return-Path: <root@IT23687882.lk>
X-Original-To: user2
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 0)
       id 17AFD21C43A4; Sun, 12 Apr 2026 04:34:24 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:34:24 +0530
To: user2@IT23687882.lk
Subject: Test
User-Agent: s-nail v14.9.22
Message-Id: <20260411230424.17AFD21C43A4@mail.IT23687882.lk>
From: root <root@IT23687882.lk>

LOCAL TEST
Return-Path: <user1@IT23687882.lk>
X-Original-To: user2@IT23687882.lk
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id C238A21C43A8; Sun, 12 Apr 2026 04:36:41 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:36:41 +0530
To: user2@IT23687882.lk
Subject: Test
User-Agent: s-nail v14.9.22
Message-Id: <20260411230641.C238A21C43A8@mail.IT23687882.lk>
From: user1@IT23687882.lk

DOMAIN TEST
Return-Path: <user1@IT23687882.lk>
X-Original-To: user2@IT23687882.lk
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id 1729621C43A8; Sun, 12 Apr 2026 04:41:31 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:41:31 +0530
To: user2@IT23687882.lk
```

Figure : Maildir test emails showing local delivery from root and user accounts.



```
user2@maduvinda7882:~/Maildir/new
maduvinda@maduvinda7882:~
Message-Id: <20260411230424.17AFD21C43A4@mail.IT23687882.lk>
From: root <root@IT23687882.lk>

LOCAL TEST
Return-Path: <user1@IT23687882.lk>
X-Original-To: user2@IT23687882.lk
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id C238A21C43A8; Sun, 12 Apr 2026 04:36:41 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:36:41 +0530
To: user2@IT23687882.lk
Subject: Test
User-Agent: s-nail v14.9.22
Message-Id: <20260411230641.C238A21C43A8@mail.IT23687882.lk>
From: user1@IT23687882.lk

DOMAIN TEST
Return-Path: <user1@IT23687882.lk>
X-Original-To: user2@IT23687882.lk
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id 1729621C43A8; Sun, 12 Apr 2026 04:41:31 +0530 (+0530)
Date: Sun, 12 Apr 2026 04:41:31 +0530
To: user2@IT23687882.lk
Subject: Test1 Mail
User-Agent: s-nail v14.9.22
Message-Id: <20260412000753.E3C3A21C4395@mail.IT23687882.lk>
From: user1@IT23687882.lk

Hello, this is a test mail from user1.
Return-Path: <user1@IT23687882.lk>
X-Original-To: user2
Delivered-To: user2@IT23687882.lk
Received: by mail.IT23687882.lk (Postfix, from userid 1001)
       id E3C3A21C4395; Sun, 12 Apr 2026 05:37:53 +0530 (+0530)
Date: Sun, 12 Apr 2026 05:37:53 +0530
To: user2@IT23687882.lk
Subject: Test Mail
User-Agent: s-nail v14.9.22
Message-Id: <20260412000753.E3C3A21C4395@mail.IT23687882.lk>
From: user1@IT23687882.lk

Hello from user1
[user2@maduvinda7882 new]$
```

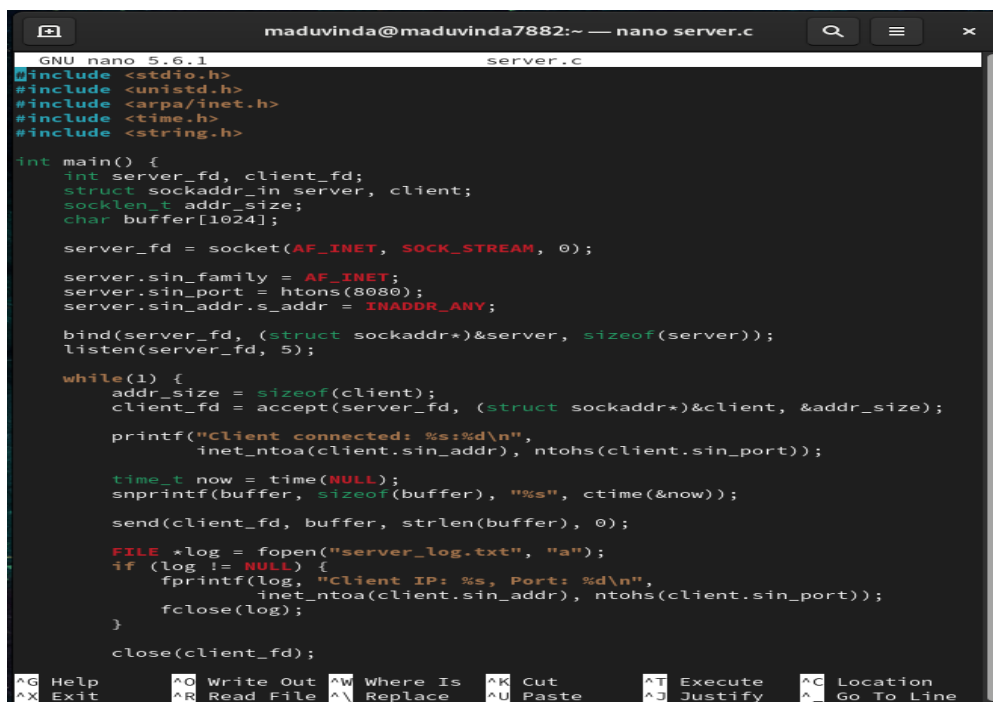
Figure : Maildir test emails confirming domain delivery and successful message exchange.

3.3. Development of Client-Server Application

A program built with C uses TCP to show how two machines exchange data across a network. Once linked, one system delivers the present date and time to the other. Instead of handling complex tasks, it focuses solely on timing information transfer.

Server

- Listening began on port 9090 once the server app finished setup. After initialization, connections from clients were accepted through that specific channel. Port assignment happened before any network activity started. Once active, it waited without delay. Network readiness followed the change in operational state
- Upon successful connection by a client, the server retrieves date and time information directly from the OS. System clock data gets captured at that moment. This timing detail comes straight from underlying platform services. Right then, temporal values are recorded without delay. Getting accurate timestamps happens automatically through internal mechanisms. At the point of access, live system time is pulled instantly. Instant contact occurs, synchronization with machine clock takes place immediately
- From there, the data travels toward the recipient. Information moves in that direction afterward. Backward flow reaches the user next. The details proceed to be returned at that stage. Transmission finishes when arrival occurs finally.
- After finishing each task, the server handles the next incoming request without overlapping. One interaction completes prior to any new one beginning. Each step follows clearly, maintaining order through repetition



```
GNU nano 5.6.1 server.c
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <time.h>
#include <string.h>

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server, client;
    socklen_t addr_size;
    char buffer[1024];

    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    server.sin_family = AF_INET;
    server.sin_port = htons(8080);
    server.sin_addr.s_addr = INADDR_ANY;

    bind(server_fd, (struct sockaddr*)&server, sizeof(server));
    listen(server_fd, 5);

    while(1) {
        addr_size = sizeof(client);
        client_fd = accept(server_fd, (struct sockaddr*)&client, &addr_size);

        printf("Client connected: %s:%d\n",
            inet_ntoa(client.sin_addr), ntohs(client.sin_port));

        time_t now = time(NULL);
        snprintf(buffer, sizeof(buffer), "%s", ctime(&now));

        send(client_fd, buffer, strlen(buffer), 0);

        FILE *log = fopen("server_log.txt", "a");
        if (log != NULL) {
            fprintf(log, "Client IP: %s, Port: %d\n",
                inet_ntoa(client.sin_addr), ntohs(client.sin_port));
            fclose(log);
        }

        close(client_fd);
    }
}
```

Figure 3.3.1.1: Server program implementation in C

```

while(1) {
    addr_size = sizeof(client);
    client_fd = accept(server_fd, (struct sockaddr*)&client, &addr_size);

    printf("Client connected: %s:%d\n",
           inet_ntoa(client.sin_addr), ntohs(client.sin_port));

    time_t now = time(NULL);
    snprintf(buffer, sizeof(buffer), "%s", ctime(&now));

    send(client_fd, buffer, strlen(buffer), 0);

    FILE *log = fopen("server_log.txt", "a");
    if (log != NULL) {
        fprintf(log, "Client IP: %s, Port: %d\n",
                inet_ntoa(client.sin_addr), ntohs(client.sin_port));
        fclose(log);
    }

    close(client_fd);
}

return 0;
}

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Figure 3.3.1.2: Server program implementation in C

Client

- The client application uses the IP address and port number of the server application to connect.
- Once the connection is successfully made, it displays the date and time received from the server application.

```

GNU nano 5.6.1 client.c
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <string.h>

int main() {
    int sock;
    struct sockaddr_in server;
    char buffer[1024];

    sock = socket(AF_INET, SOCK_STREAM, 0);

    server.sin_family = AF_INET;
    server.sin_port = htons(8080);
    server.sin_addr.s_addr = inet_addr("127.0.0.1");

    connect(sock, (struct sockaddr*)&server, sizeof(server));

    recv(sock, buffer, sizeof(buffer), 0);
    printf("Server Time: %s\n", buffer);

    FILE *log = fopen("client_log.txt", "a");
    if (log != NULL) {
        fprintf(log, "Connected to server (IP: 127.0.0.1, Port: 8080)\n");
        fclose(log);
    }

    close(sock);

    return 0;
}

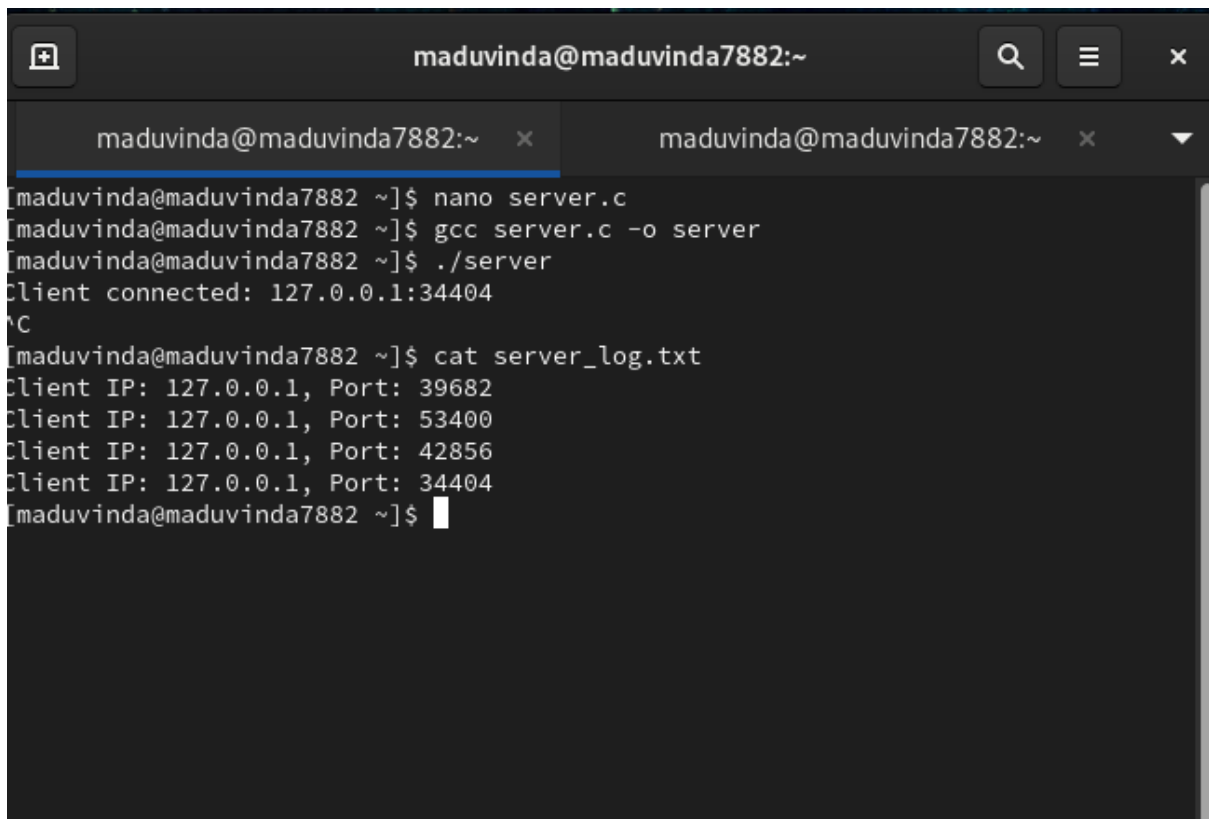
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

Figure 3.3.2.1: Client program implementation in C

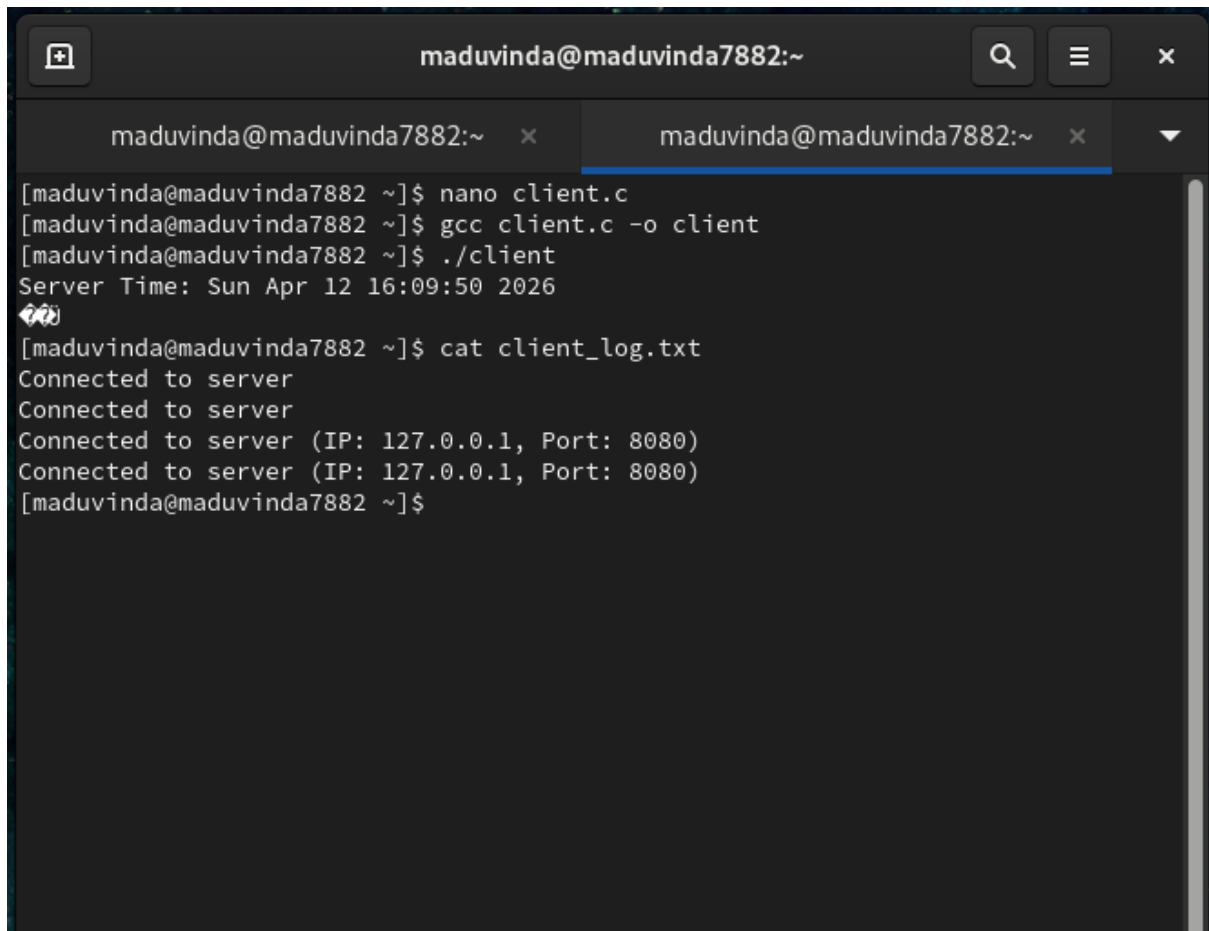
Testing:

- Server Application was run in the CentOS Virtual Machine, whereas Client Application was run in the Fedora (or CentOS) Virtual Machine. Client Application successfully established connection with the Server and showed the current date and time, proving the communication between client and server was working properly.

A terminal window titled 'maduvinda@maduvinda7882:~' with a search icon, menu icon, and close button. It contains two tabs, both labeled 'maduvinda@maduvinda7882:~'. The terminal output shows the following commands and results:

```
[maduvinda@maduvinda7882 ~]$ nano server.c
[maduvinda@maduvinda7882 ~]$ gcc server.c -o server
[maduvinda@maduvinda7882 ~]$ ./server
Client connected: 127.0.0.1:34404
^C
[maduvinda@maduvinda7882 ~]$ cat server_log.txt
Client IP: 127.0.0.1, Port: 39682
Client IP: 127.0.0.1, Port: 53400
Client IP: 127.0.0.1, Port: 42856
Client IP: 127.0.0.1, Port: 34404
[maduvinda@maduvinda7882 ~]$
```

Figure 3.3.3: Server running on CentOS and logging client socket IP/port details.



The image shows a terminal window with a dark background. The title bar at the top reads "maduvinda@maduvinda7882:~". Below the title bar, there are two tabs, both labeled "maduvinda@maduvinda7882:~". The terminal content shows the following sequence of commands and output:

```
[maduvinda@maduvinda7882 ~]$ nano client.c
[maduvinda@maduvinda7882 ~]$ gcc client.c -o client
[maduvinda@maduvinda7882 ~]$ ./client
Server Time: Sun Apr 12 16:09:50 2026
🔊🔊
[maduvinda@maduvinda7882 ~]$ cat client_log.txt
Connected to server
Connected to server
Connected to server (IP: 127.0.0.1, Port: 8080)
Connected to server (IP: 127.0.0.1, Port: 8080)
[maduvinda@maduvinda7882 ~]$
```

Figure 3.3.4: Client program execution recording socket IP and port details.

4.Conclusion

As a result, the present task was successfully finished. It clearly showed how to create a network, configure necessary services, and develop a working client-server application. Setting up DNS and Mail Servers in the Linux operating system was proven to be effective and productive.

Also, developing a TCP client-server application allowed me to gain additional knowledge about socket programming and network interaction. Adding logging capabilities made monitoring and controlling communication more manageable.

In summary, this assignment helped me get better knowledge of network setup, configuring services, and developing a client-server system based on TCP/IP.

5.References

- CentOS Stream 9 Official Documentation
- BIND9 Administration Manual
- Postfix Mail System Guide and Setup Tutorials
- W. Richard Stevens – UNIX Network Programming